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MATHEMATICS

STANDARD SEVEN

Primary School Leaving Examination (PSLE) 2021 Answer Key and Worked Solutions

Subject	Mathematics
Code	04E
Level	Standard Seven
Year	2021
Exam Board	NECTA
Exam	Primary School Leaving Examination (PSLE)
Type	Answer key and worked solutions
Questions	45 questions

Try each question before checking the solution. Follow the steps, compare your working, and practise the parts you find difficult.

Source paper: Mathematics-S7-2021.pdf

Section A — Multiple choice

1. Which is smallest: 0.457, $\frac{2}{3}$, 40%, 0.53 or 0.80?

Step 1: Convert to decimals: $\frac{2}{3} = 0.666\dots$, and $40\% = 0.40$.

Step 2: Compare 0.457, 0.666..., 0.40, 0.53 and 0.80. The least is 0.40.

Final Answer: C — 40%.

2. Which drink is preferred most? Fanta XCI, Pepsi XC, tea CIX, water CVI and milk CX.

Step 1: Read the Roman numbers: XCI = 91, XC = 90, CIX = 109, CVI = 106 and CX = 110.

Step 2: The greatest number is 110, which belongs to milk.

Final Answer: E — Milk.

3. State the place value of the underlined 2 in 504251.

Step 1: Read places from the right: 1 ones, 5 tens, 2 hundreds, 4 thousands, 0 ten thousands and 5 hundred thousands.

Step 2: The digit 2 is in the hundreds column; its numerical value is 200.

Final Answer: D — Hundreds.

4. A full 10-litre gas jar has used 1.5 litres. What fraction remains?

Step 1: Subtract the amount used: $10 - 1.5 = 8.5$ litres.

Step 2: Fraction remaining = $\frac{8.5}{10} = \frac{85}{100} = \frac{17}{20}$.

Final Answer: C — $\frac{17}{20}$.

5. Of 60 pupils attending and 20 absent, what percentage did not attend?

Step 1: Total class = $60 + 20 = 80$ pupils.

Step 2: Absent percentage = $(\frac{20}{80}) \times 100\% = 25\%$.

Final Answer: C — 25%.

6. Round 3,845,308 cattle to the nearest ten thousand.

Step 1: The ten-thousands digit is 4. The next digit, in the thousands place, is 5.

Step 2: Because the next digit is at least 5, increase 4 to 5 and replace all following digits with zeros.

Final Answer: D — 3,850,000.

7. Mwanakwetu was two in 1961. Give her age in 2020 in Roman numerals.

Step 1: Elapsed years = $2020 - 1961 = 59$.

Step 2: Add her original age: $59 + 2 = 61$. Write 61 as $50 + 10 + 1 = L + X + I$.

Final Answer: B — LXI (61 years).

8. Identify the primes among 21, 29, 37, 45, 49, 51, 53, 57, 59 and 61.

Step 1: A prime has exactly two positive factors: 1 and itself.

Step 2: Remove $21 = 3 \times 7$, $45 = 5 \times 9$, $49 = 7^2$, $51 = 3 \times 17$ and $57 = 3 \times 19$.

Step 3: The remaining numbers have no smaller prime divisor up to their square roots.

Final Answer: C — 29, 37, 53, 59 and 61.

9. Complete the ages: 13, 18, 23, __.

Step 1: Find the increase: $18 - 13 = 5$ and $23 - 18 = 5$.

Step 2: Continue by adding 5: $23 + 5 = 28$.

Final Answer: A — 28 years.

10. Continue the revenue sequence 2, 4, 7, 12, 19, __.

Step 1: Find successive increases: 2, 3, 5 and 7.

Step 2: These are consecutive prime numbers. The next prime is 11.

Step 3: Add it to the last term: $19 + 11 = 30$.

Final Answer: E — 30.

Study Tip: Look at the differences between terms before deciding how a sequence continues.

11. Complete the weights: 59, 55, 51, __, 43, 39, 35.

Step 1: Each successive weight decreases by 4.

Step 2: The missing value is $51 - 4 = 47$. Check: $47 - 4 = 43$.

Final Answer: B — 47.

12. Complete the five-day attendance sequence: 78, 75, 72, 69, __.

Step 1: Attendance falls by 3 each day: $78 - 75 = 75 - 72 = 72 - 69 = 3$.

Step 2: Fifth-day attendance = $69 - 3 = 66$.

Final Answer: C — 66 pupils.

Study Tip: The final digits in the scan are faded; NECTA's analysis confirms this sequence.

13. Villagers raised sh 325,500. Each contributed sh 500. How many contributed?

Step 1: Number of contributors = total contribution $\div$ contribution per person.

Step 2: $325,500 \div 500 = 651$. Check: $651 \times 500 = 325,500$.

Final Answer: D — 651 villagers.

14. Find the years between MCMLXI and MCMLXXX, and express the difference in Roman numerals.

Step 1: $MCMLXI = 1000 + 900 + 50 + 10 + 1 = 1961$; $MCMLXXX = 1980$.

Step 2: Difference = $1980 - 1961 = 19$.

Step 3: Write 19 as $10 + 9$: $X + IX = XIX$.

Final Answer: B — XIX (19 years).

15. Find the sum represented by the two abacuses in the paper.

Step 1: Read the first abacus by place value: 275,256. Read the second as 5,025.

Step 2: Add by columns: $275,256 + 5,025 = 280,281$.

Step 3: Carry 1 from the ones into tens, and carry 1 from thousands into ten thousands.

Final Answer: A — 280,281.

16. John made sh 600,000 profit. Each of 300 chickens earned sh 1,500; each egg tray earned sh 1,000. How many trays were sold?

Step 1: Chicken profit = $300 \times 1,500 = \text{sh } 450,000$.

Step 2: Egg profit = $600,000 - 450,000 = \text{sh } 150,000$.

Step 3: Trays sold = $150,000 \div 1,000 = 150$.

Final Answer: D — 150 trays.

17. Ashura has 43 mangoes and Asha 17. How many should Ashura give Asha so both have equal numbers?

Step 1: Together they have $43 + 17 = 60$ mangoes; equal shares are $60 \div 2 = 30$.

Step 2: Asha needs $30 - 17 = 13$. Ashura then has $43 - 13 = 30$.

Final Answer: B — 13 mangoes.

Study Tip: A transfer changes both totals, so divide the original difference by two.

18. Jerry has 208 workers, each paid sh 55,460 a day. Find the daily total.

Step 1: Multiply the number of workers by each daily wage.

Step 2: $55,460 \times 208 = (55,460 \times 200) + (55,460 \times 8)$.

Step 3: $11,092,000 + 443,680 = 11,535,680$.

Final Answer: E — sh 11,535,680.

19. Find the difference between 998,999 and 819,937 votes.

Step 1: Align digits by place value and subtract: $998,999 - 819,937$.

Step 2: The difference is 179,062. Check by adding: $819,937 + 179,062 = 998,999$.

Final Answer: C — 179,062 votes.

20. A class has 480 pupils; $\frac{7}{12}$ are girls. Find the difference between girls and boys.

Step 1: Girls = $(\frac{7}{12}) \times 480 = 280$.

Step 2: Boys = $480 - 280 = 200$.

Step 3: Difference = $280 - 200 = 80$.

Final Answer: D — 80 pupils.

21. Of sh 3,600,000, Juma gets $\frac{1}{5}$, Ally $\frac{4}{9}$, and Rehema the rest. What is the largest share?

Step 1: Juma receives $3,600,000 \div 5 = \text{sh } 720,000$.

Step 2: Ally receives $(3,600,000 \div 9) \times 4 = \text{sh } 1,600,000$.

Step 3: Rehema receives $3,600,000 - 720,000 - 1,600,000 = \text{sh } 1,280,000$.

Step 4: Compare all three shares: Ally's is largest.

Final Answer: D — sh 1,600,000.

22. Of 1,560 cattle, 35% are calves and 10% bulls. How many are cows?

Step 1: Cow percentage = $100\% - 35\% - 10\% = 55\%$.

Step 2: Cows = $(55/100) \times 1,560 = 858$.

Final Answer: A — 858 cows.

23. Farida has sh 550,000. She buys a TV for 162,000, camera for 45,700, couches for 87,900, ring for 68,500, and three pairs of shoes at 20,000 each. Find the balance.

Step 1: Shoes cost $3 \times 20,000 = \text{sh } 60,000$.

Step 2: Total spending = $162,000 + 45,700 + 87,900 + 68,500 + 60,000 = \text{sh } 424,100$.

Step 3: Balance = $550,000 - 424,100 = \text{sh } 125,900$.

Final Answer: E — sh 125,900.

24. A baby sleeps at 7:40 a.m. for seven hours. When does he wake?

Step 1: In 24-hour time the starting time is 07:40.

Step 2: $07:40 + 7 \text{ hours} = 14:40$. Convert back: $14 - 12 = 2 \text{ p.m.}$

Final Answer: A — 2:40 p.m.

25. Write 9:45 a.m. in the 24-hour system.

Step 1: Morning hours keep their value in 24-hour time.

Step 2: Write 9 with a leading zero, followed by the minutes 45.

Final Answer: C — 0945 hours (09:45).

26. Convert a visit lasting 840 hours into days.

Step 1: One day has 24 hours.

Step 2: Days = $840 \div 24 = 35$.

Final Answer: E — 35 days.

27. Yusufu arrives at 8:30 a.m., thirty minutes late. When should he have arrived?

Step 1: Subtract the delay from the actual arrival time.

Step 2: 8:30 a.m. – 30 minutes = 8:00 a.m.

Final Answer: D — 8:00 a.m.

28. John, Anna and Suzan receive profits of sh 30,000, 60,000 and 90,000. Each gives $\frac{1}{3}$ to a hospital. Find the donation.

Step 1: Combine the profits: $30,000 + 60,000 + 90,000 = \text{sh } 180,000$.

Step 2: Donation = $(\frac{1}{3}) \times 180,000 = \text{sh } 60,000$.

Step 3: Equivalently add individual donations $10,000 + 20,000 + 30,000$.

Final Answer: A — sh 60,000.

29. How many lines of symmetry does the capital-I-shaped figure in the paper have?

Step 1: Reflect the figure across a vertical line through its centre; its left and right halves match.

Step 2: Reflect it across a horizontal centre line; top and bottom match.

Step 3: A diagonal reflection does not match the long middle stem and short end bars.

Final Answer: B — 2 lines of symmetry.

30. A cubic water well has volume 64 m³. Find one side.

Step 1: For a cube, volume = side × side × side = s^3 .

Step 2: Since $4 \times 4 \times 4 = 64$, $s = 4$.

Final Answer: B — 4 m.

31. A tyre has diameter 100 cm. How many turns cover 471 m? Use $\pi = 3.14$.

Step 1: Convert diameter: 100 cm = 1 m.

Step 2: Distance per turn = circumference = $\pi d = 3.14 \times 1 = 3.14$ m.

Step 3: Turns = $471 \div 3.14 = 150$.

Final Answer: D — 150 complete turns.

32. Name the type of angle EFG shown in the roof diagram.

Step 1: The middle letter F identifies the vertex.

Step 2: The opening at F is greater than 90° and less than 180° , which defines an obtuse angle.

Final Answer: B — Obtuse angle.

33. A climber stops at 3,955 m on a 5,895 m mountain. How far below the summit is she?

Step 1: Remaining height = summit height – height reached.

Step 2: $5,895 - 3,955 = 1,940$.

Final Answer: A — 1,940 m.

34. Mama Shamba puts 200.25 litres of oil into 267 equal bottles. Find each bottle's capacity.

Step 1: Capacity per bottle = $200.25 \div 267 = 0.75$ litres.

Step 2: Convert litres to millilitres: $0.75 \times 1,000 = 750$ ml.

Final Answer: D — 750 millilitres.

35. Mpanda travels 72 km in 15 minutes. Find speed in metres per second.

Step 1: Convert distance: $72 \text{ km} = 72,000 \text{ m}$. Convert time: $15 \times 60 = 900 \text{ s}$.

Step 2: Speed = distance $\div$ time = $72,000 \div 900 = 80$.

Final Answer: A — 80 m/s.

36. Choose the correct order for masses 3,000 mg, 230 cg, 0.5 dg, 0.39 dag and 2.4 g.

Step 1: Convert to grams: $3,000 \text{ mg} = 3 \text{ g}$; $230 \text{ cg} = 2.3 \text{ g}$; $0.5 \text{ dg} = 0.05 \text{ g}$; $0.39 \text{ dag} = 3.9 \text{ g}$.

Step 2: The descending values are 3.9, 3, 2.4, 2.3, 0.05 g.

Step 3: Only option D follows that consistent order.

Final Answer: D — 0.39 dag, 3,000 mg, 2.4 g, 230 cg, 0.5 dg.

37. A number minus nine equals one quarter of that number. Find the number.

Step 1: Let the number be x : $x - 9 = x/4$.

Step 2: Multiply by 4: $4x - 36 = x$. Therefore $3x = 36$.

Step 3: Divide by 3: $x = 12$. Check: $12 - 9 = 3 = 12/4$.

Final Answer: A — 12.

38. Simplify $2(3m - 2n + 5m)$.

Step 1: Collect like terms inside the brackets: $3m + 5m = 8m$.

Step 2: Multiply both terms by 2: $2(8m - 2n) = 16m - 4n$.

Final Answer: C — $16m - 4n$.

Study Tip: Keep the minus sign when multiplying $-2n$ by 2; the scan's sign is faint.

39. Crop sales are cashews 0.3, rice $2/5$, maize 0.1 and the remaining millet. Find the millet pie-chart angle.

Step 1: Convert rice to decimal: $2/5 = 0.4$.

Step 2: Millet proportion = $1 - (0.3 + 0.4 + 0.1) = 0.2$.

Step 3: Multiply by a full turn: $0.2 \times 360^\circ = 72^\circ$.

Final Answer: E — 72° .

40. Four children receive $4x$, $2x$, $3x$ and x shillings. The third receives sh 28,800. Find the original total.

Step 1: From $3x = 28,800$, divide by 3: $x = 9,600$.

Step 2: Total shares = $(4 + 2 + 3 + 1)x = 10x$.

Step 3: $10 \times 9,600 = \text{sh } 96,000$.

Final Answer: B — sh 96,000.

Section Summary: Convert units before calculating, keep place values aligned, and use the whole population as the percentage denominator. For money problems, identify the required operation before choosing an option.

Section B — Written solutions

41. List the odd perfect squares between 0 and 50.

Step 1: Squares in the range are 0, 1, 4, 9, 16, 25, 36 and 49.

Step 2: Keep only numbers not divisible by 2: 1, 9, 25 and 49.

Final Answer: 1, 9, 25 and 49.

42. A right angle is split into $(2y - 10)^\circ$, $(4y + 20)^\circ$ and $(40 - 2y)^\circ$. Find y .

Step 1: A right angle totals 90° : $(2y - 10) + (4y + 20) + (40 - 2y) = 90$.

Step 2: Collect terms: $4y + 50 = 90$; then $4y = 40$.

Step 3: Divide by 4: $y = 10$. Check the angles: $10^\circ + 60^\circ + 20^\circ = 90^\circ$.

Final Answer: $y = 10$.

43. Find the volume of a book box measuring 26 cm by 9 cm by 12 cm.

Step 1: Volume of a cuboid = length $\times$ width $\times$ height.

Step 2: $26 \times 9 \times 12 = 234 \times 12 = 2,808$.

Step 3: All lengths are in centimetres, so use cubic centimetres.

Final Answer: 2,808 cm³.

44. Two adults and eight children pay sh 48,000 in total. A child pays half an adult's fare. Find one child's fare.

Step 1: Let a child's fare be c , so an adult pays $2c$.

Step 2: Total = $2(2c) + 8c = 12c = 48,000$.

Step 3: $c = 48,000 \div 12 = 4,000$. Check: adults 16,000 plus children 32,000 = 48,000.

Final Answer: sh 4,000 per child.

45. Annual fees are sh 44,665.35 for one pupil. How much is collected from 32 pupils?

Step 1: Multiply the fee by 32 = 30 + 2.

Step 2: $44,665.35 \times 30 = 1,339,960.50$; $44,665.35 \times 2 = 89,330.70$.

Step 3: Add: $1,339,960.50 + 89,330.70 = 1,429,291.20$.

Final Answer: sh 1,429,291.20.

Section Summary: Show the equation or formula first, substitute the given values, then calculate and include units. Check angle sums and substitute solutions back into equations.

Source Checks

NECTA PSLE 2021 Mathematics performance analysis